Practice workbook · Adding up an arithmetic series between two limits

Evaluating a sigma sum that does not start at r = 1 by subtracting two sums from r = 1, or by finding the first term, last term and number of terms.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-05

1. Question 3 · part b · 2 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part b only.

3 (a) Show that $\sum_{r=1}^{n}(5r-3)=\frac{n}{2}(5n-1)$ (3)

(b) Hence, or otherwise, evaluate $\sum_{r=31}^{60}(5r-3)$ (2)

Given that $\sum_{r=1}^{n}(5r-3)=3783$

(c) find the value of n (3)

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P 7 6 5 0 6 A 0 6 3 2

1. Question 3 · part b · 2 marks · Working and answer

Working and answer

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2. Question 5 · part b · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part b only.

5 (a) Show that $\sum_{r=1}^{n}(3r+5) = \frac{1}{2}n(3n+13)$ (3)

(b) Hence evaluate $\sum_{r=35}^{50}(3r+5)$ (2)

Given that $\sum_{r=1}^{n}(3r+5) = 385$

(c) find the value of n. (3)

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2. Question 5 · part b · 2 marks · Working and answer

Working and answer

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3. Question 1 · part b · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part b only.

1 (a) Show that $\sum_{r=1}^{n}(3r+2) = \frac{n}{2}(3n+7)$

(3)

(b) Hence, or otherwise, evaluate $\sum_{r=10}^{40}(3r+2)$

(2)

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(Total for Question 1 is 5 marks)

3. Question 1 · part b · 2 marks · Working and answer

Working and answer

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4. Question 9 · part b · 3 marks

2021 NOV · Paper 01

Further practice · Attempt part b only.

9 (a) Show that $\sum_{r=1}^{n}(5r-1)=\frac{n}{2}(3+5n)$ (3)

(b) Hence, or otherwise, evaluate $\sum_{r=10}^{20}(5r-1)$ (3)

The sum of the first n terms of an arithmetic series is S_n where $S_n=\sum_{r=1}^{n}(5r-1)$

The rth term of this series is u_r

Given that $S_n=12u_{n+1}+52$

(c) find the value of n (5)

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4. Question 9 · part b · 3 marks · Working and answer

Working and answer

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5. Question 2 · part b · 4 marks

2024 NOV · Paper 01

Further practice · Attempt part b only.

2 The sum of the fifth, sixth and seventh terms of an arithmetic series A is nine times the sum of the first and second terms.
The third term of A is 12

(a) Find the first term and common difference of A

(5)

The nth term of A is u_n

(b) Find the value of $\sum_{r=15}^{60} u_r$

(4)

The sum to n terms of A is S_n

Given that $2S_n - 5u_n = 10$

(c) find the value of n

(4)

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5. Question 2 · part b · 4 marks · Working and answer

Working and answer

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6. Question 2 · part b · 4 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

2 The nth term of an arithmetic series is u_n

Given that $u_5 = 46$ and that $u_{20} = 181$

(a) find

(i) the common difference of this series,

(ii) the first term of this series.

(4)

(b) Evaluate $\sum_{n=21}^{30} u_n$

(4)

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6. Question 2 · part b · 4 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 2 continued

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(Total for Question 2 is 8 marks)

6. Question 2 · part b · 4 marks · Working and answer

Working and answer

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7. Question 4 · part b.i · 2 marks

2025 NOV · Paper 01

Further practice · Attempt part b.i only.

4 Given that $\sum_{r=1}^{n}(4r+A)=\frac{n}{2}(4n-6)$ where A is a constant

(a) show that $A=-5$

(3)

Hence, or otherwise,

(b) (i) evaluate $\sum_{21}^{60}(4r-5)$

(2)

(ii) find the greatest value of n such that

$$\sum_{r=1}^{n}(4r-5)<3418$$

(3)

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7. Question 4 · part b.i · 2 marks · Working and answer

Working and answer

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